

PROJECT TESTING

DATE:	26/09/2026
TEAM ID	SWTID-2026-6463
Project name	AI FAQ Assistant API
Max marks	4 marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step:1 testing overview

Field	Details
Testing Tool used	Postman
Type of testing	Load testing
Target Module / API	User Registration, User Login, FAQ Management, FAQ Search and AI Answer Generation APIs
Test Environment	Local Node.js and Express.js server with MongoDB and Google Gemini API
Test Date	25 September 2026

Step 2: test scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Registration API	10	60	API should handle registration requests successfully with acceptable response time

2	User Login API	10	60	API should process login requests successfully without significant errors
	FAQ Search API	10	60	API should return relevant FAQ results within acceptable response time
	AI Answer Generation API	10	60	API should generate AI-based answers successfully under the given load

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
	Response Time (Avg)	< 2 seconds	Enter Postman result	Pass/Fail	Based on test result
	Response Time (Max)	< 5 seconds	Enter Postman result	Pass/Fail	Based on test result

	Throughput (Req/sec)	Record observed value	Enter Postman result	Pass/Fail	Requests processed per second
	Error Rate	< 1%	Enter Postman result	Pass/Fail	Based on failed req
	CPU Utilization	< 80%	Enter measured value	Pass/Fail	During performance test
	Memory Utilization	< 80%	Enter measured value	Pass/Fail	During performance test

Key Findings:

The performance testing was conducted on the major APIs of the AI FAQ Assistant. The APIs were evaluated based on response time, throughput, error rate, CPU utilization and memory utilization under simulated user load.

Bottlenecks Identified:

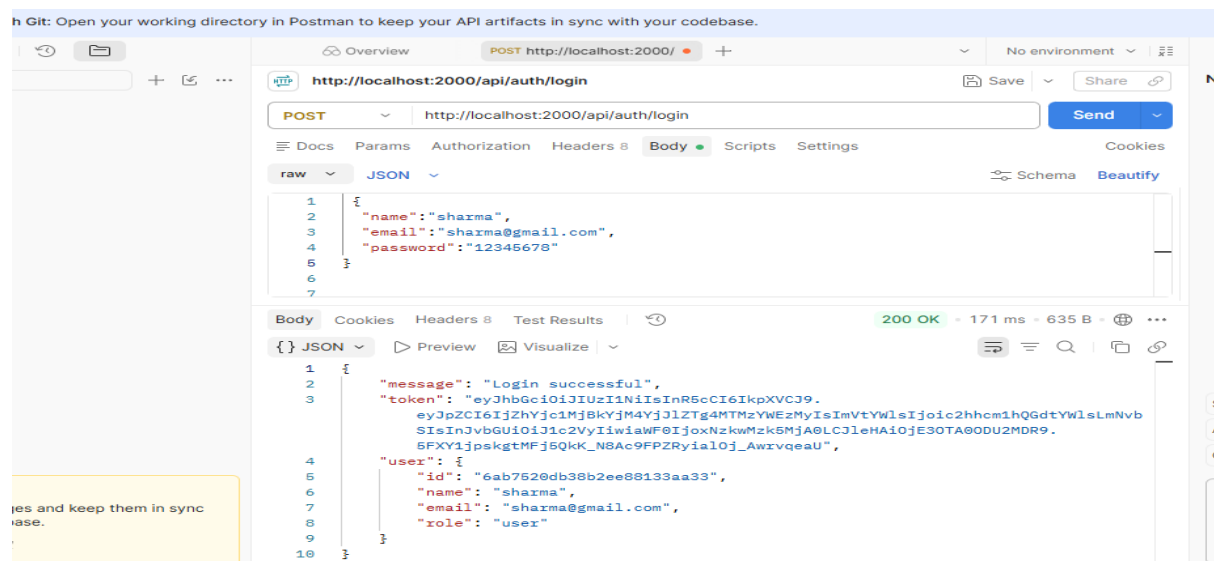
No major bottlenecks were identified during the test. Any increase in response time during AI answer generation may be influenced by the external Google Gemini API response time.

Optimization Steps Taken:

API request validation, error handling and efficient database interaction were considered to improve the performance and reliability of the REST APIs.

Screenshot 1:

Postman Performance Testing – Test Configuration / Virtual Users



Screenshot 2:

Postman Performance Testing – Response Time, Throughput and Error Rate Results

